

INVITATION TO TENDER FOR NOMINATED SUB-CONTRACT WORKS FOR SUPPLY AND INSTALLATION OF EXTRA-LOW VOLTAGE ELECTRICAL WORKS TO PROPOSED DEVELOPMENT OF AN INTEGRATED ACUTE AND COMMUNITY HOSPITAL AT BEDOK NORTH, SINGAPORE

APPENDIX U: AI-BASED SMART BUILDING TECHNOLOGIES

The scope and the performance specifications below set out the minimum requirement and serve as guidelines. The tenderers shall submit their proposed systems for the S.O. approval.

1. AI-Based Chiller Plant Efficiency Optimisation System

1.1. AI-Driven Energy Optimisation

The AI-powered system shall be cable of continuously collect the key parameters of the chilled water system, power consumption, weather data etc, carry out self-training by using the past data and provide real time setting adjustment to optimise chiller plant performance.

- **Real-time load balancing:** Adjusts chiller sequencing based on cooling demand.
- **Dynamic setpoint optimisation:** Regulates chilled water and condenser water temperatures setpoints.
- **AI-based control algorithms:** Predict and adjust system settings based on occupancy and weather patterns.

1.3. Data Analytics & Reporting

- **Automated trend analysis** for energy consumption and performance benchmarking.
- **Custom dashboards and reports** for facility managers to track system efficiency.
- **AI-driven recommendations** for system upgrades and efficiency improvements.

1.4. Communication & Control Layer

- **IoT Gateway:** Collects data either from IoT sensors or BMS sensors.
- **Cloud based AI Engine:** Processes real-time data, self-training and optimises chiller operation using cloud which is fully complied with the cybersecurity requirements. Alternatively, the Sub-Contractor may subscribe to Healthcare Commercial Cloude managed by Synapxe.

1.5. AI-Based Control & Optimisation Layer

- **Machine Learning Algorithms:** Predict cooling demand and optimise chiller operation and efficiency.
- **Neural Networks or Equivalent:** Self-learning AI models that adapt to seasonal changes and occupancy patterns.

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- **Automated Setpoint Control:** Adjusts setpoints of water temperatures, pump speeds and cooling tower fan operations.

1.6. Visualization & User Interface

- **Web-based Dashboard** for real-time monitoring.
- **Mobile App Integration** for remote access.
- **AI-Generated Insights & Reports** for decision-making.

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2. AI-Based Predictive Maintenance for Critical MEP Equipment

2.1. AI-Powered Predictive Maintenance Capabilities

AI-driven predictive maintenance shall be capable of using **real-time data, machine learning, and IoT sensors** to:

- **Continuous equipment monitoring**
- **Detect anomalies** in equipment performance.
- **Predict failures and send alerts** before they occur.
- **Optimize maintenance schedules** to minimize disruptions.
- **Reduce unplanned downtime** and enhance patient safety.

2.2. Critical MEP Equipment

The AI-based system shall monitor and manage the following critical MEP equipment:

Category	Equipment
HVAC Systems	Chillers Chilled Water Pumps Condenser Water Pumps Cooling Towers Heat Pumps Heat Recovery Units AHUs/FCUs/Fans serving Operating Theatres, Emergency Department, Intensive Care Units (ICU), High Dependency (HD) Wards, Isolation Wards, Integrated Control Centre (ICC), Fire Command Centre (FCC), Security Command Centre, Laboratories, Pharmacies, Blood Storage Facility, Radiology Departments, Material Management Department
Power Systems	UPS Generators Main HT/LT Electrical Panels
Medical Gas Systems	Vacuum Pumps Air Compressors AGSS Pumps
Plumbing & Water Supply	Pumps Ejectors
Fire Protection	Fire Pumps

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2.3. AI Analytics & Predictive Maintenance Engine

- **Machine Learning Algorithms:** Analyze sensor data to detect failure patterns.
- **Neural Networks & Deep Learning:** Self-learning models improve prediction accuracy over time.
- **Anomaly Detection & Alerts:** Identifies abnormal behavior and triggers early warnings.
- **Remaining Useful Life (RUL) Estimation:** Predicts how long equipment can operate before requiring maintenance.

2.4. Control & User Interface

- **Web-Based Dashboard:** Displays real-time health status of MEP systems.
- **Mobile App for Facility Teams:** Provides remote access to maintenance insights.
- **Automated Reports & Maintenance Logs:** AI-generated insights for performance tracking.